

Communicating with Data – Visual Encoding 2

Dr. Ab Mosca (they/them)

Sponsored by the Center for the Environment, Ecological Design and Sustainability, Environmental Science and Policy Program, Smith Office for the Arts, Lazarus Center for Career Development, Program in Statistical & Data Sciences, Sustainable Capital Programs, Endowed Lecture Fund, and the departments of Computer Science, Mathematical Sciences, and East Asian Languages and Cultures.

From Computer Science to Center Stage

A CONVERSATION WITH VIENNA TENG

Acclaimed singer-songwriter, performing artist, and creative entrepreneur Vienna Teng will lead an inspiring conversation about her journey from studying computer science and working at Cisco Systems, a leading technology company, to pursuing a full-time music career.



MONDAY, SEPT. 21

12-1:30 P.M.

McCartney Hall 106

REGISTER



THE SMITH OFFICE FOR THE ARTS PRESENTS:

MUSIC FOR DESSERT:

SONGS & CHAT
WITH
VIENNA TENG!

SEPT 21 - 7:30PM
ALUMNAE GYM 3RD FLOOR

VIENNA TENG IS A SONGWRITER, PERFORMER AND ORGANIZER. SHE HAS BEEN FEATURED ON NPR AND THE LATE SHOW, RECORDED ALBUMS FOR ROUNDER/UNIVERSAL RECORDS, AND HER MUSIC GARNERS OVER 450,000 ANNUAL LISTENERS ON STREAMING SERVICES.

- HEAR HER PLAY A FEW SONGS (POSSIBLY JOIN IN)
- CHAT ABOUT HER MUSIC CAREER
- EAT DESSERT

SmithArts



START WITH JOY

A CLIMATE ACTION WORKSHOP with Vienna Teng

Join Vienna Teng, acclaimed singer-songwriter, former McKinsey Sustainability fellow and one of the Grist 50 "climate innovators to watch," for the afternoon!

Rather than focusing on the climate crisis itself, Vienna will help you reflect on what gives you energy and meaning (start with joy), identify whom you could join in taking action (stop being one person), integrate climate solutions into your busy life (stack with other stuff), and choose one thing and take a simple first step (stress less). Come play with sticky notes, enjoy delicious food, hear a little music - and take your place in the epic, unfinished story of climate action.



MONDAY, SEPT. 21

4:30-6:30 P.M.

CEEDS, Wright Hall 005

REGISTER



CEEDS
Center for the Environment,
Ecological Design & Sustainability

SMITHIES IN
SDS

SUMMER SHARE-OUT



lunch provided for
the first 25 people

Tuesday, September 22nd

12:15pm

Sabin-Reed 301

Share about your summer or just
come to hear about your peers and
professors! Send a google slide
with photos to

smithiesinsds@smith.edu

(doesn't have to be sds related!)



join the
SSDS club
Discord!



Plan for Today

- Recap:
 - Visualization building blocks
 - Mapping data to visuals
- More complex examples
- Practice

Recap

Find a small group to work with. Answer the following:

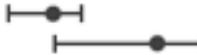
1. What 3 types of marks that make up visualizations?
2. What is the difference between a magnitude and an identity visual channel?
3. Name 6 magnitude visual channels.
4. Name 3 identity visual channels.

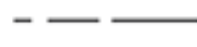
- Remember... **Big idea behind visualization**
 - Visual Encoding:** Map data dimensions to visual dimensions in a principled way

Recap

➔ **Magnitude Channels: Ordered Attributes**

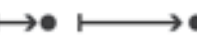
Position on common scale 

Position on unaligned scale 

Length (1D size) 

Tilt/angle 

Area (2D size) 

Depth (3D position) 

Color luminance 

Color saturation 

Curvature 

Volume (3D size) 

➔ **Identity Channels: Categorical Attributes**

Spatial region 

Color hue 

Motion 

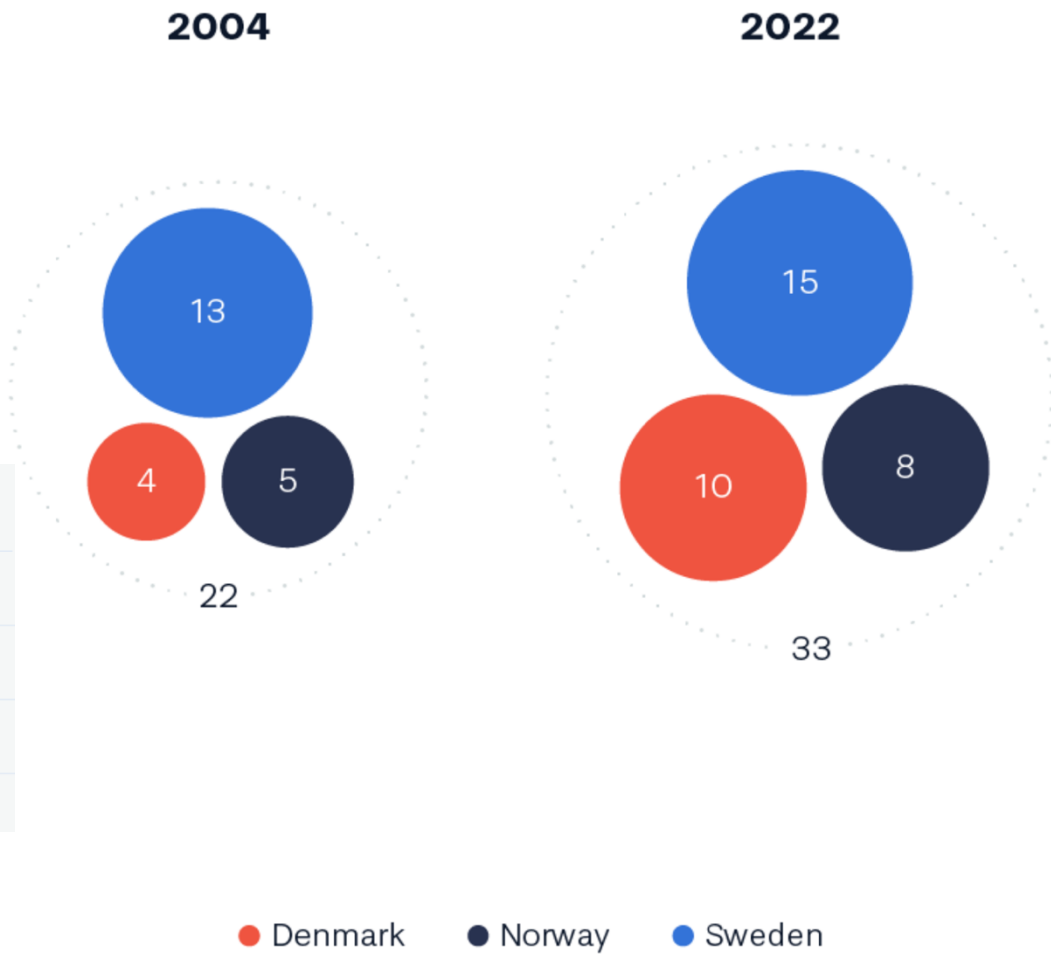
Shape 

Recap

Find a small group to work with. Answer the following:

1. What marks and channels compose this vis? What is the visual encoding?

Number of World Heritage Sites			
	Norway	Denmark	Sweden
2004	5	4	13
2022	8	10	15

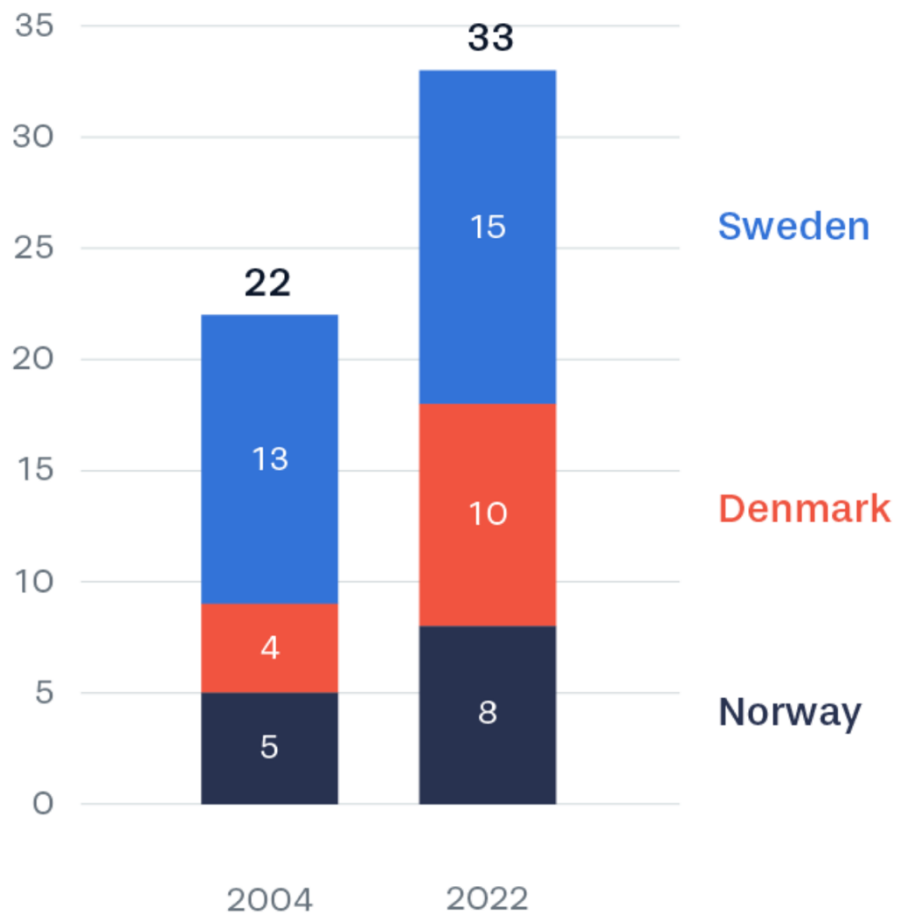


Recap

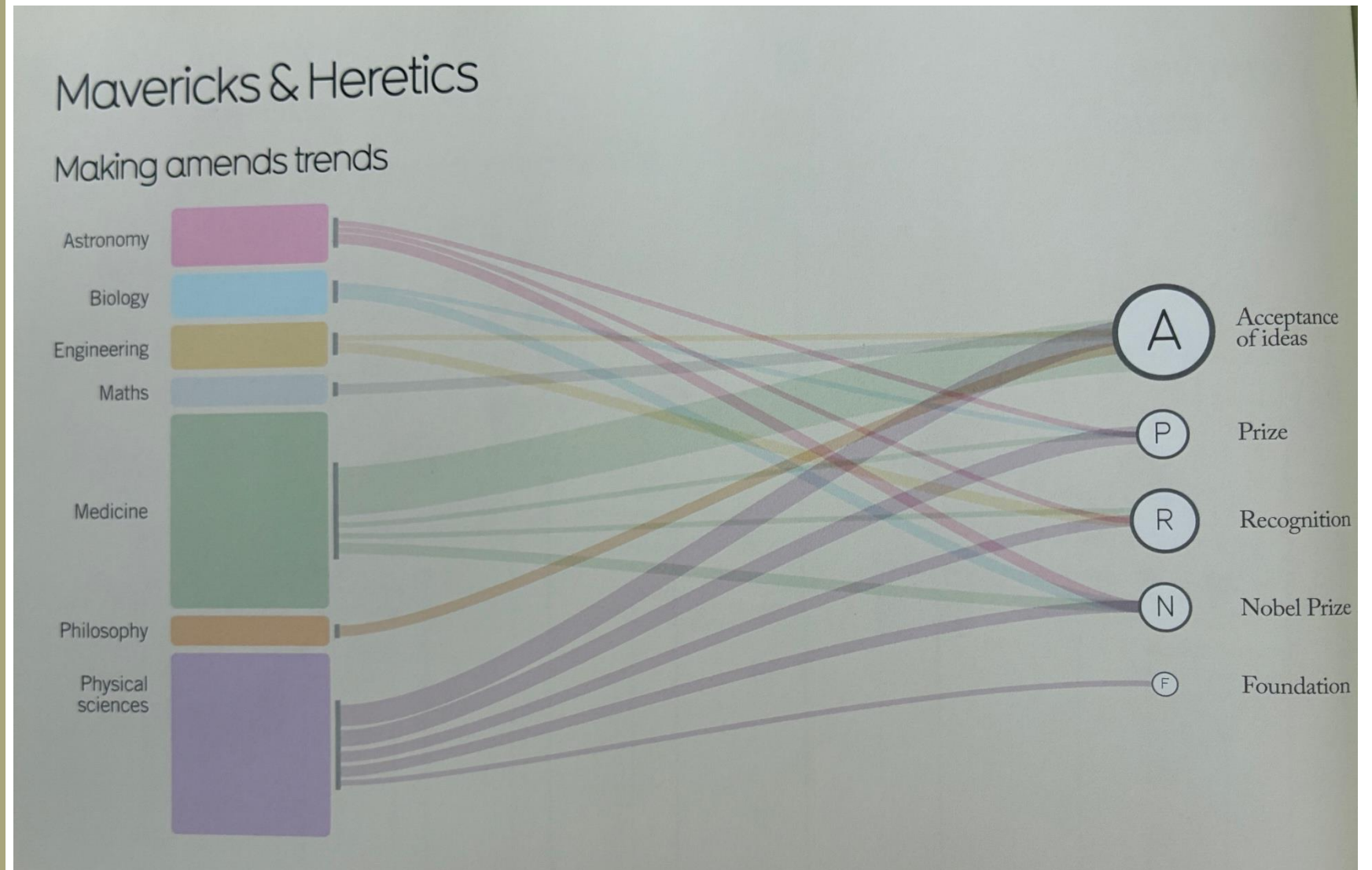
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Visual Encoding Examples

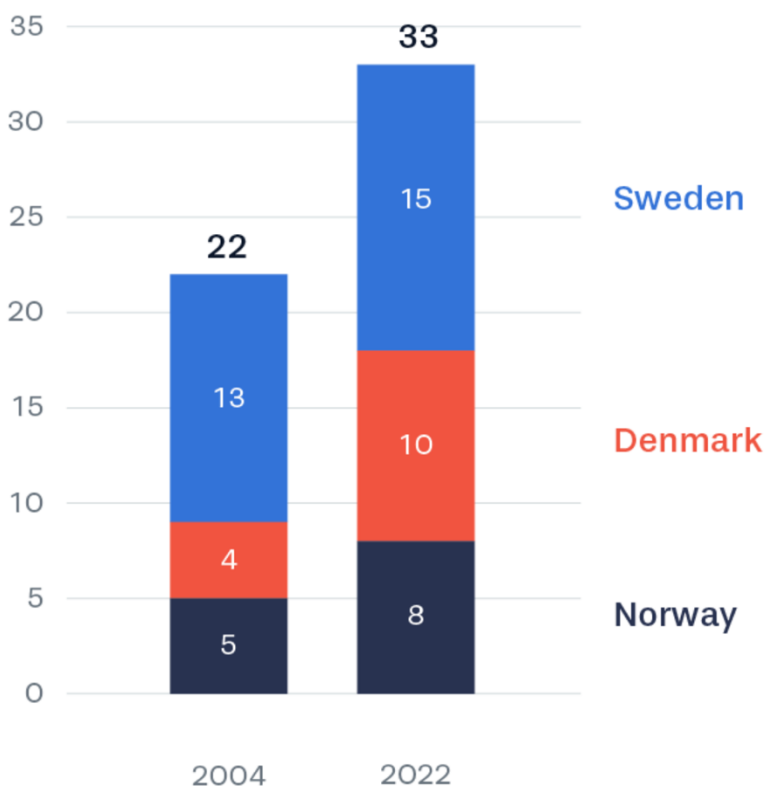
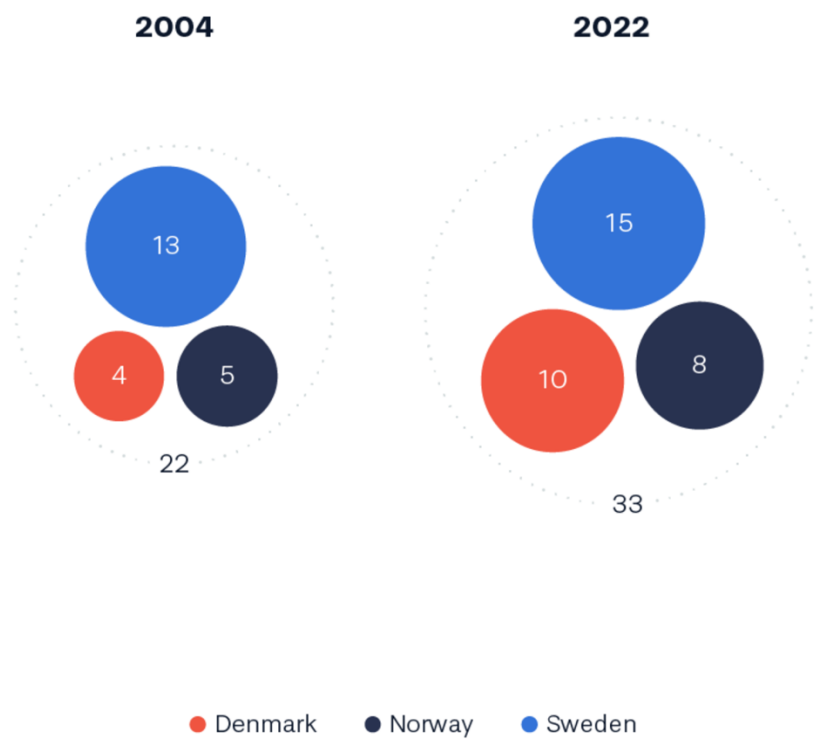


McCandless, D. (2014). *Knowledge is beautiful: Impossible ideas, invisible patterns, hidden connections—visualized*. HarperCollins.

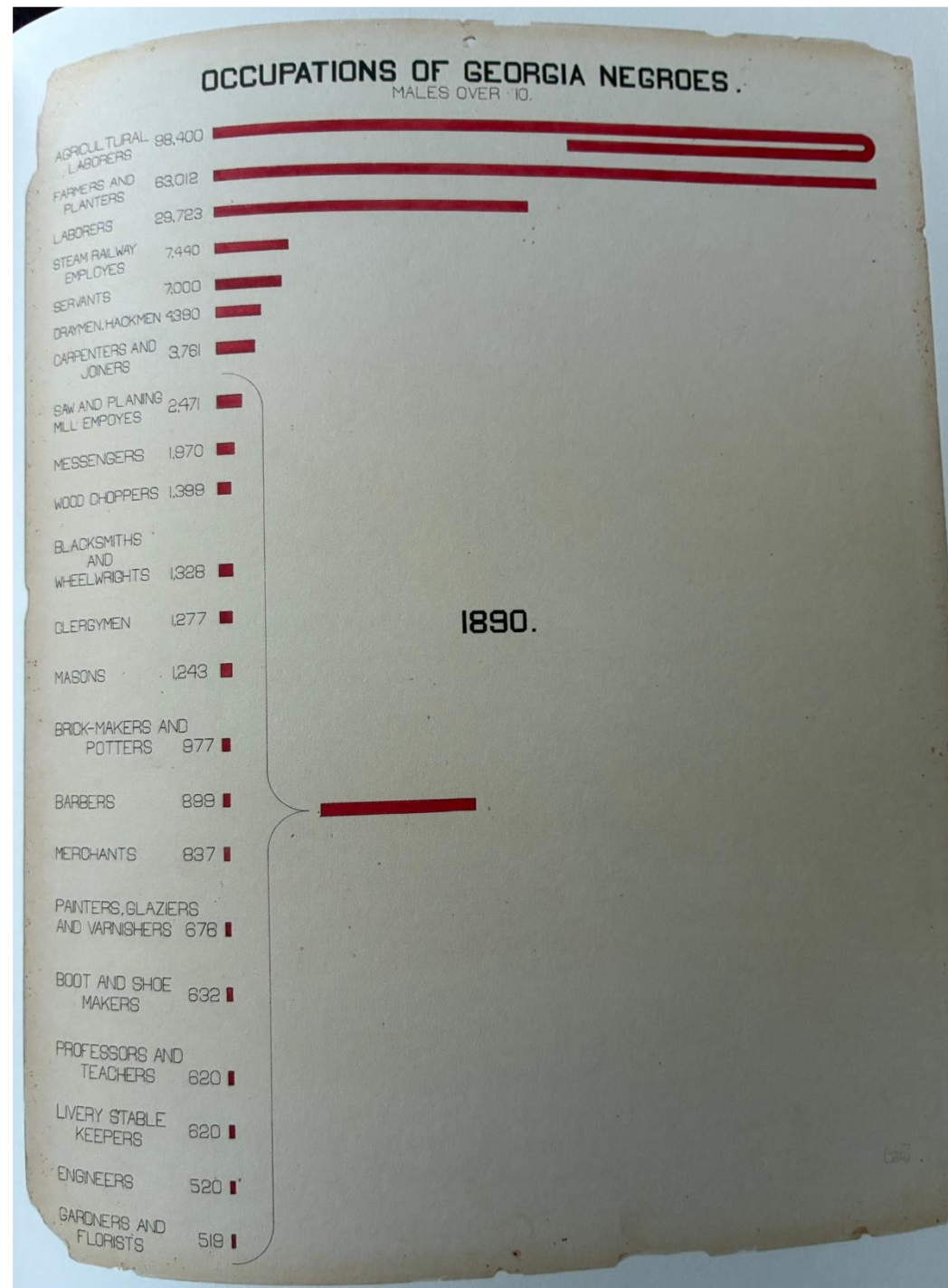
Recap

Find a small group to work with. Answer the following:

1. Which is better? Why?



Visual Encoding Examples

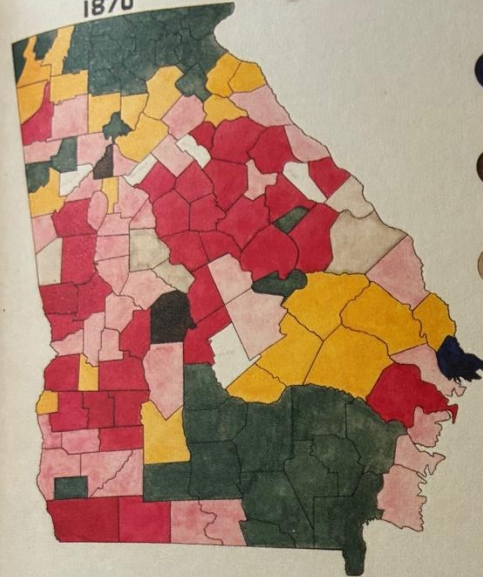


Battle-Baptiste, W., & Rusert, B. (Eds.). (2018).
*W. E. B. Du Bois's data portraits: Visualizing
Black America*. Princeton Architectural Press.

Visual Encoding Examples

NEGRO POPULATION OF GEORGIA BY COUNTIES.

1870



● BETWEEN 20,000 AND 30,000

● 15,000 TO 20,000

● 10,000 TO 15,000

1880



● 5,000 TO 10,000

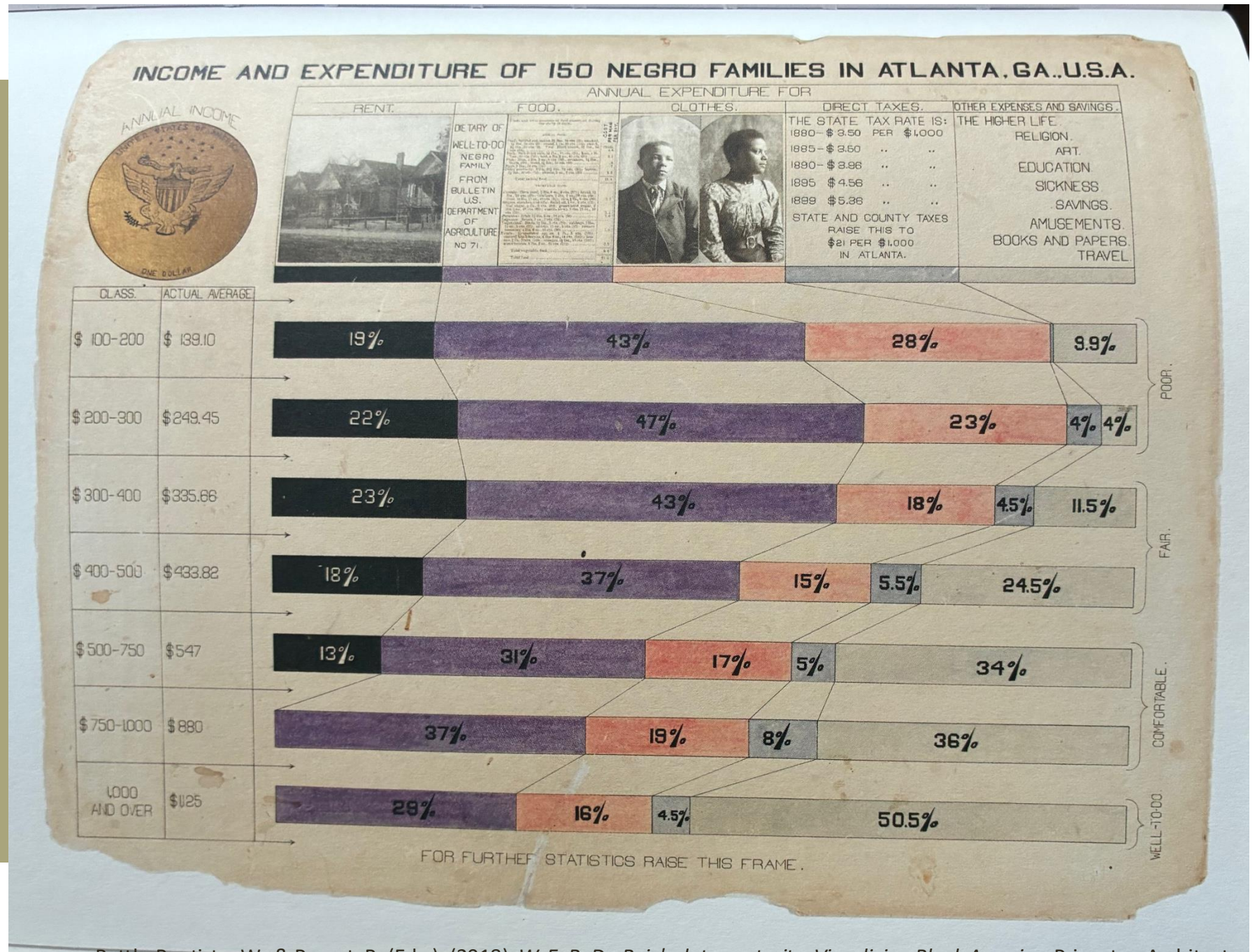
● 2,500 TO 5,000

● 1,000 TO 2,500

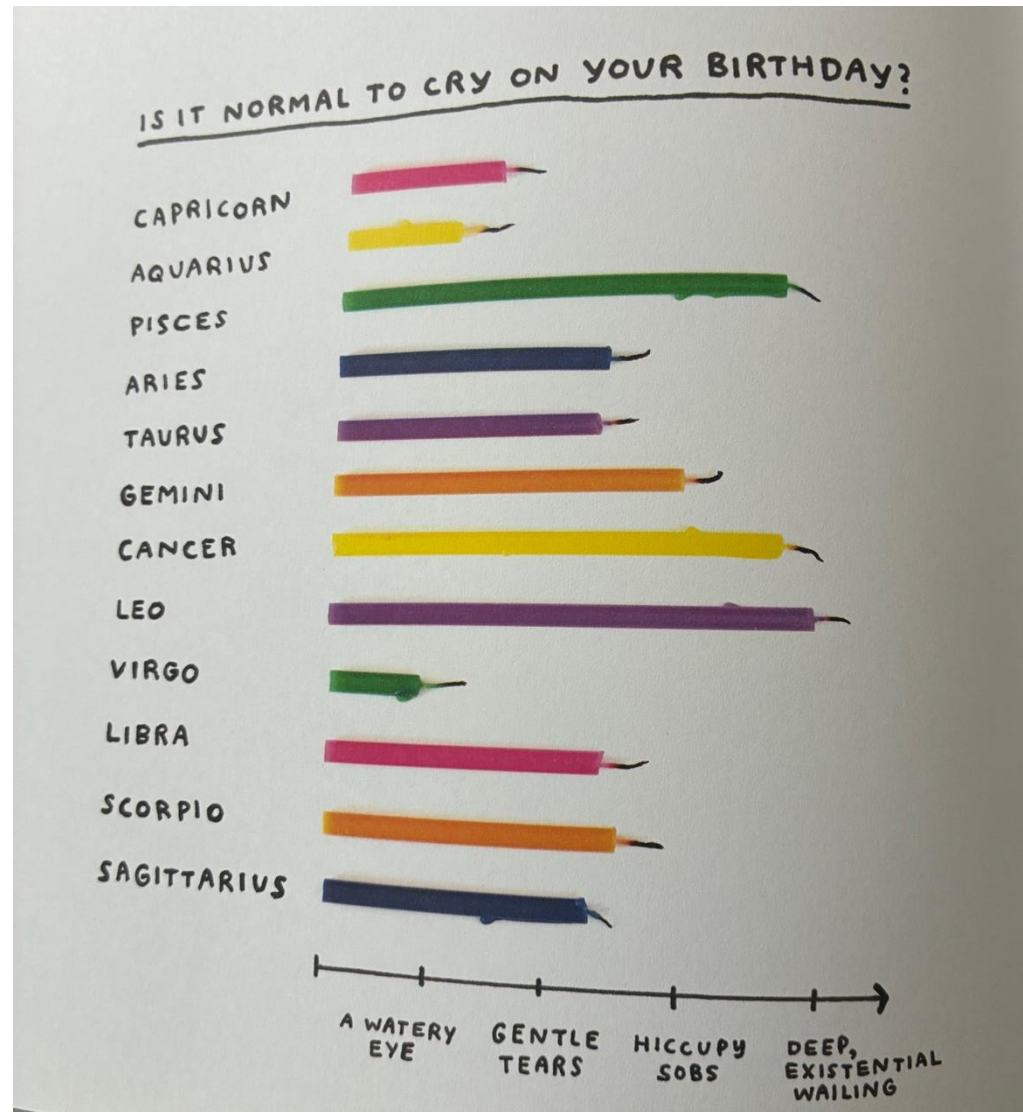
● UNDER 1,000

Battle-Baptiste, W., & Rusert, B. (Eds.). (2018).
*W. E. B. Du Bois's data portraits: Visualizing
Black America*. Princeton Architectural Press.

Visual Encoding Examples

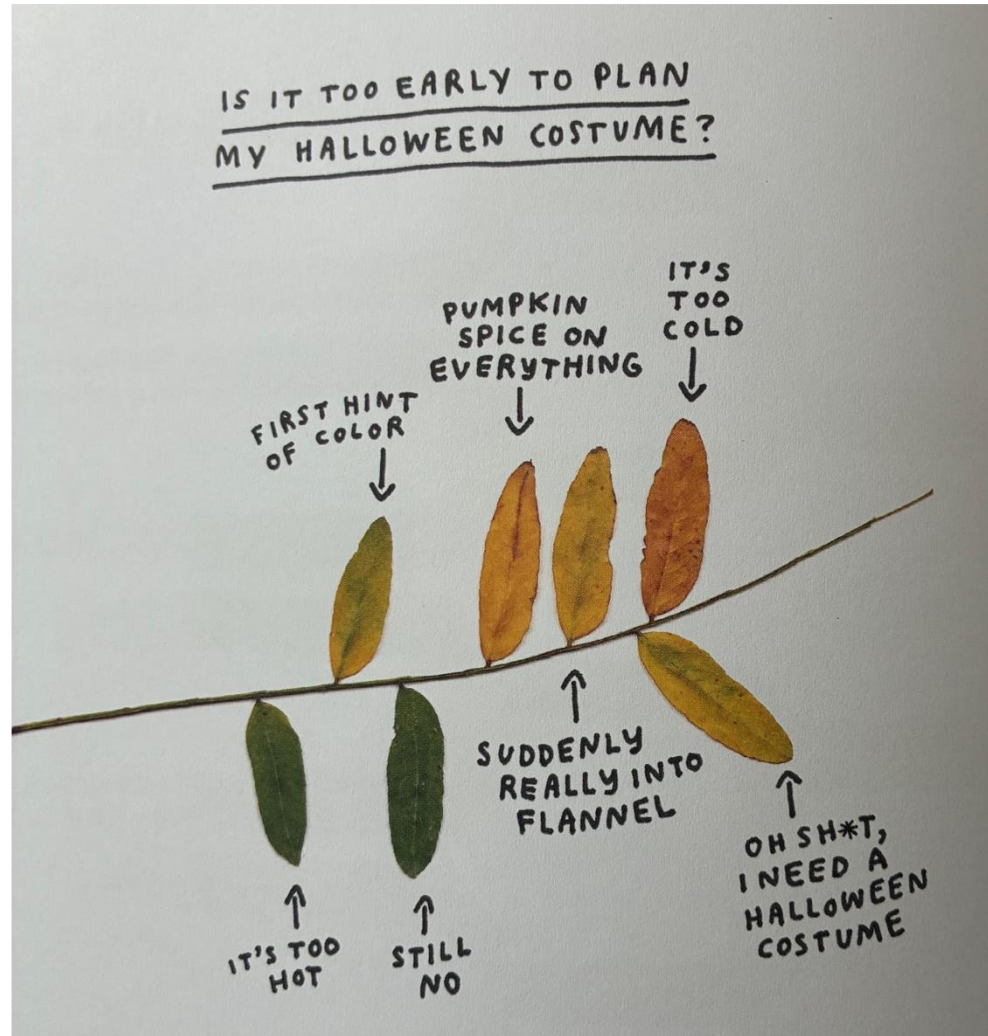


Visual Encoding Examples

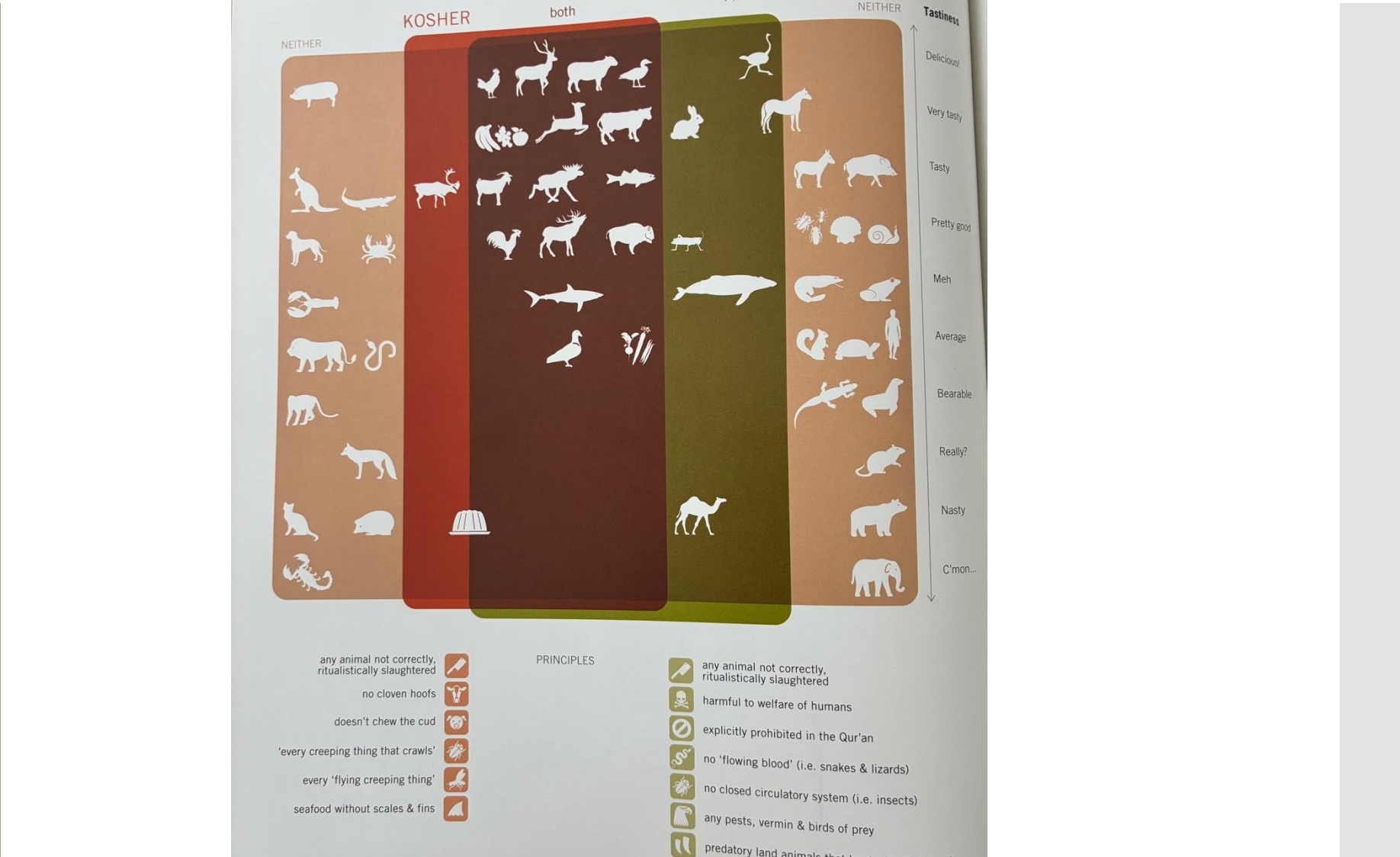


Rial, M. (2019). *Am I overthinking this? Over-answering life's questions in 101 charts*

Visual Encoding Examples

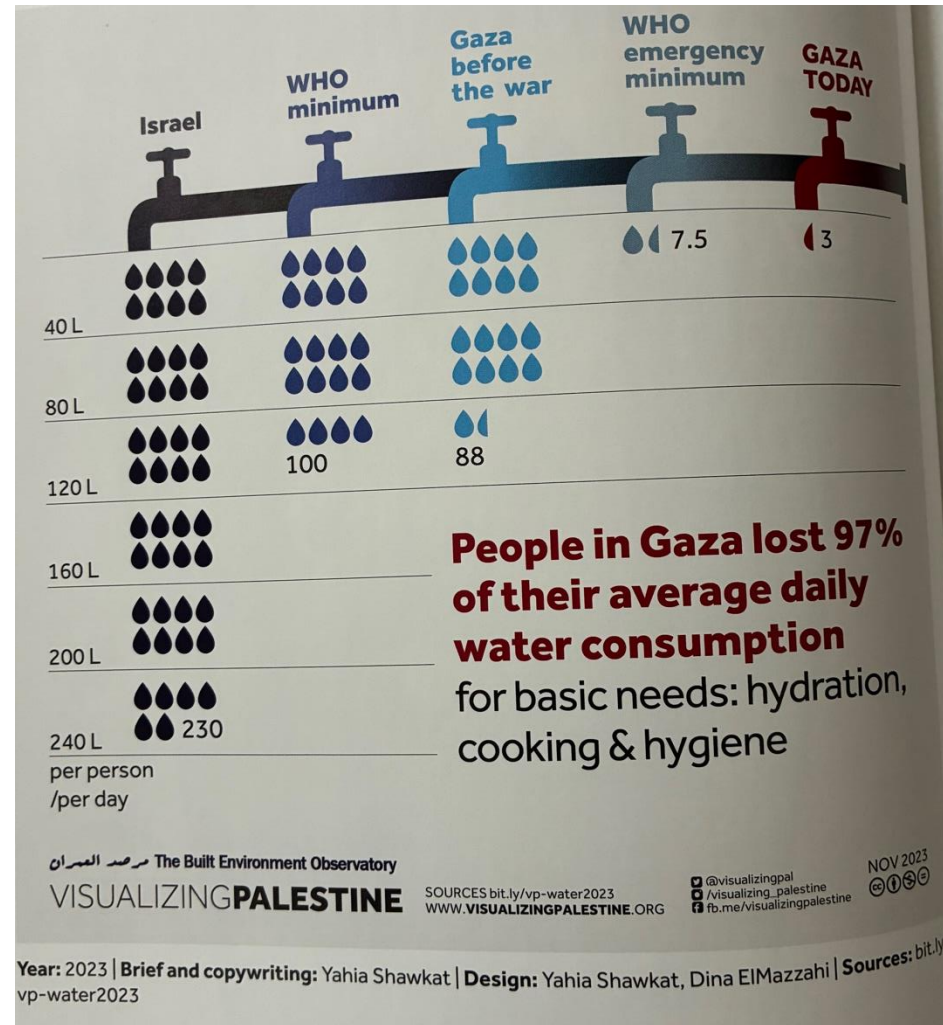


Visual Encoding Examples



McCandless, D. (2014). *Knowledge is beautiful: Impossible ideas, invisible patterns, hidden connections—visualized*. HarperCollins.

Visual Encoding Examples



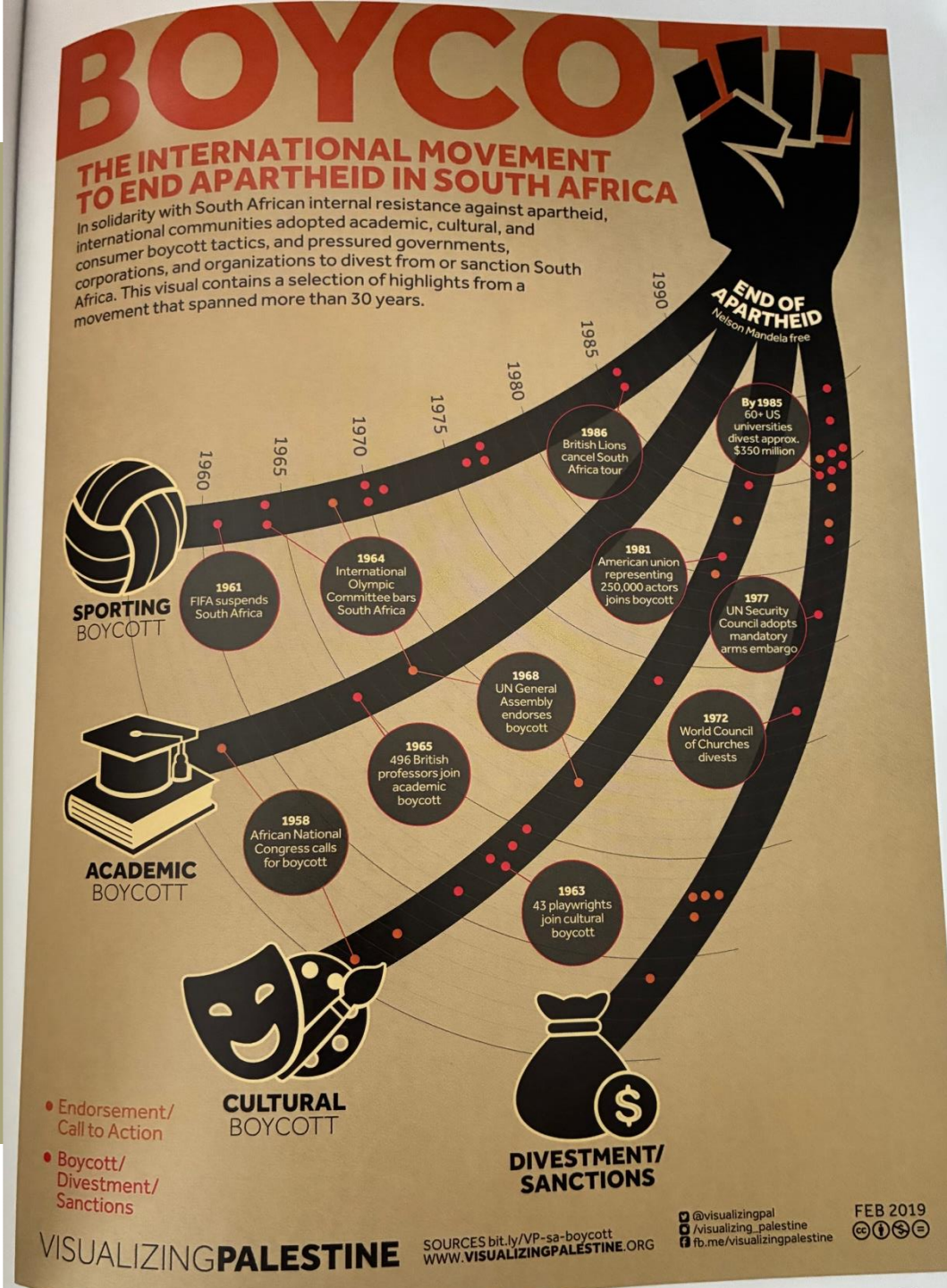
Anderson, J., Batarseh, A., & El Gazzar, Y. (Eds.). (2024). *Visualizing Palestine: A Chronicle of Colonialism and the Struggle for Liberation*. Haymarket Books

Visual Encoding Examples



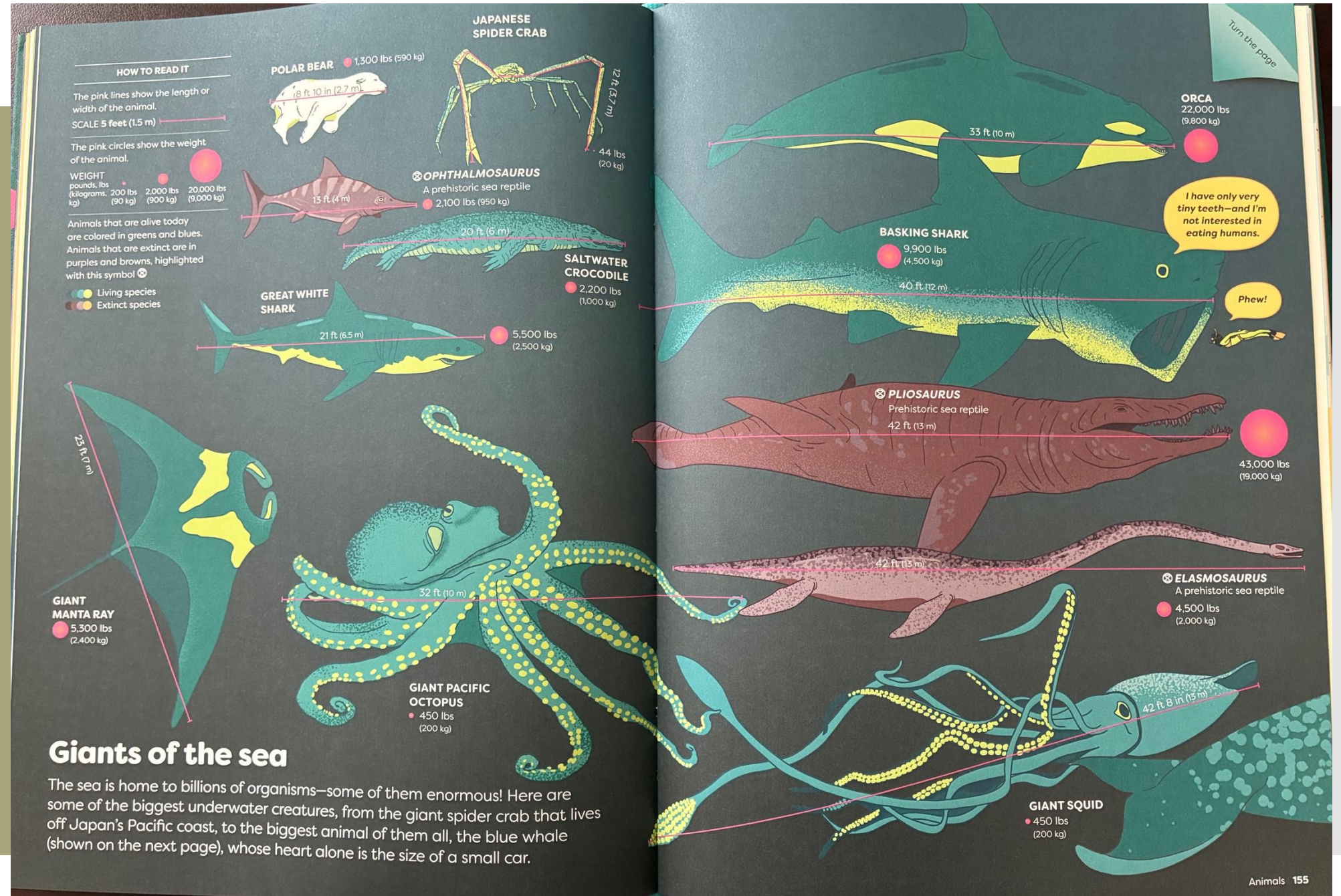
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Visual Encoding Examples



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Visual Encoding Examples



Visual Encoding Examples

The strongest animals on Earth

Which animal is stronger, an elephant or an ant? The elephant can carry the heavier weight, of course. But perhaps a fairer way to decide is to measure how much weight each animal can carry relative to the weight of its own body. Looking at it this way, small but powerful animals, such as ants and beetles, are even stronger—relative to the size of their bodies—than gorillas, bears, and elephants!



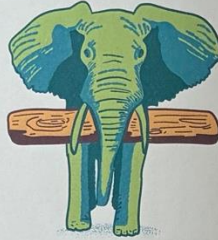
HARPY EAGLE

The harpy eagle can carry 17 lbs (7.5 kg), the same as **(1 time)** its own weight.



GRIZZLY BEAR

A grizzly bear can carry 550 lbs (250 kg), the same as **(1 time)** its own weight.



AFRICAN ELEPHANT

This elephant can carry 20,000 lbs (9,000 kg), **1.5 times** its own weight of 13,000 lbs (6,000 kg).



TIGER

A tiger can carry 880 lbs (400 kg), twice **(2 times)** its own weight of 440 lbs (200 kg).



MUSK OX

A musk ox can carry 1,250 lbs (570 kg), twice **(2 times)** its own weight of 630 lbs (285 kg).

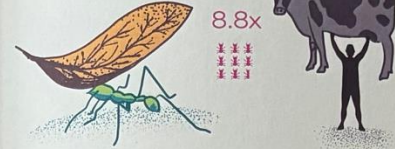
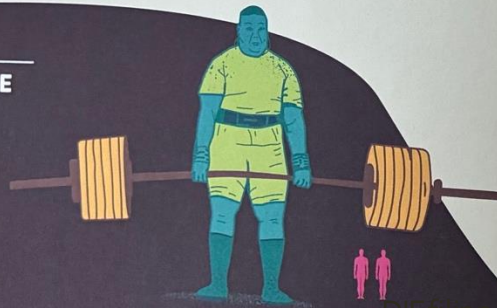


GORILLA

A gorilla can carry 1,400 lbs (640 kg), **4 times** its own weight of 350 lbs (160 kg).

HOW STRONG ARE HUMANS?

The world record-holding weightlifter **Tamara Walcott** can lift 641 lbs (291 kg), twice **(2 times)** her own weight of 295 lbs (134 kg).



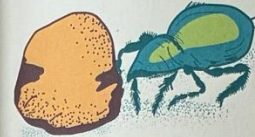
LEAFCUTTER ANT

This remarkable ant can carry **nearly 9 times** its own weight of 0.0007 oz (0.02 g). This is equivalent to a **human carrying a cow!**



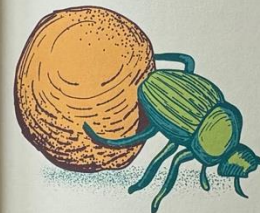
RHINOCEROS BEETLE

A rhinoceros beetle can carry **100 times** its own weight of 0.08 oz (2.4 g). This is equivalent to a **human carrying an elephant!**



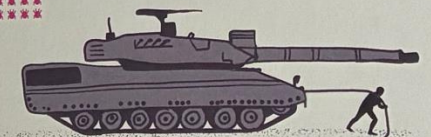
ORIBATID MITE

This mite can pull **530 times** its own weight of 0.0000035 oz (0.0001 g). This is equivalent to a **human pulling the weight of a Brachiosaurus dinosaur!**



DUNG BEETLE

A dung beetle can pull **more than 1,100 times** its own weight of 0.003 oz (0.09 g). This is equivalent to a **human pulling a tank!**



Visual Encoding Examples

Which animals kill the most humans?

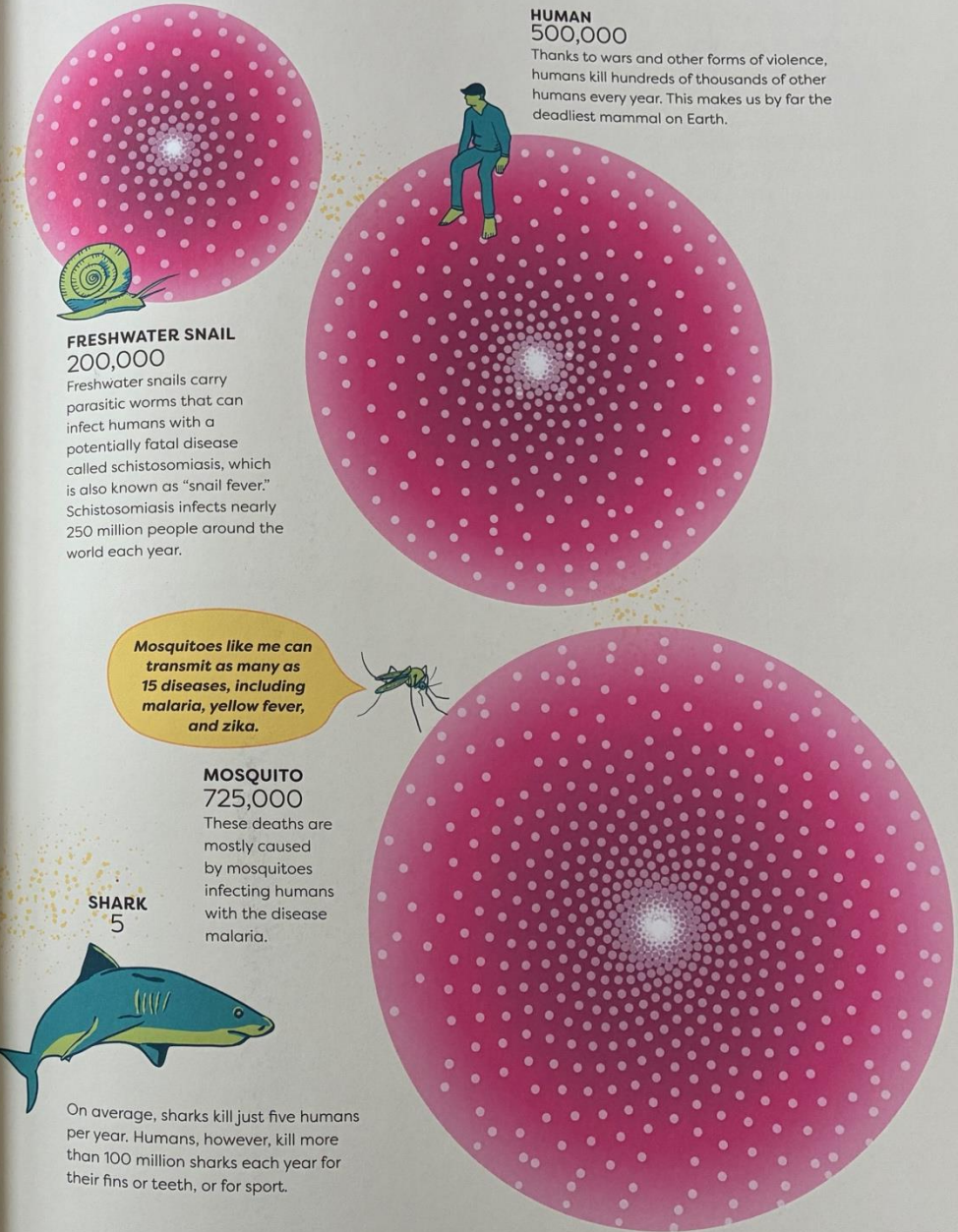
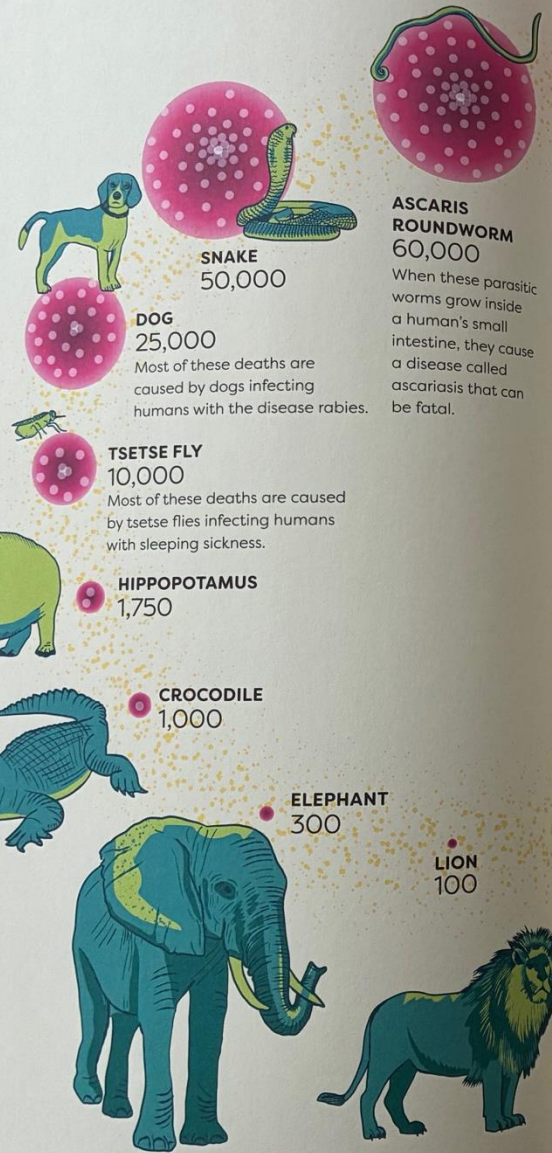
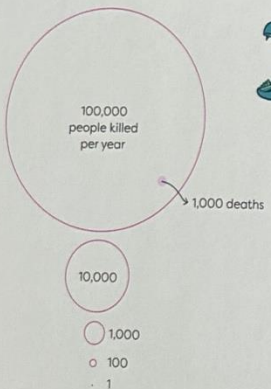
Which animal do you think poses the greatest danger to humans? If you said a hungry shark or a poisonous spider, you would be... completely wrong! Because the deadliest animal for humans is the tiny mosquito.

WHY ARE MOSQUITOES SO DEADLY?

Mosquitoes do not kill humans directly. But they do transmit dangerous diseases between humans and other animals when they bite them and drink their blood. The deadliest of these diseases is malaria, which is found only in certain parts of the world but has killed millions of people throughout human history.

HOW TO READ IT

The size of the pink circles shows the estimated number of humans that each animal species kills per year.



Mosquitoes like me can transmit as many as 15 diseases, including malaria, yellow fever, and Zika.

MOSQUITO
725,000
These deaths are mostly caused by mosquitoes infecting humans with the disease malaria.

SHARK
5
On average, sharks kill just five humans per year. Humans, however, kill more than 100 million sharks each year for their fins or teeth, or for sport.

HUMAN
500,000
Thanks to wars and other forms of violence, humans kill hundreds of thousands of other humans every year. This makes us by far the deadliest mammal on Earth.

D'Elipippo, Valentina, Andrew Pettie, and Conrad Quilty-Harper. *Britannica's Encyclopedia Infographica: 1,000s of Facts & Figures About Earth, Space, Animals, the Body, Technology, & More Revealed in Pictures*. Britannica Books, 2023.

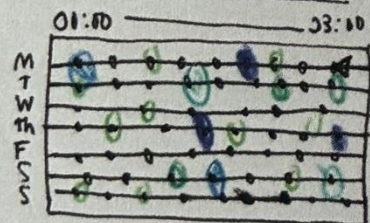
DEAR DATA - WEEK 12

A WEEK OF PEOPLE

THE DATA: I TRACKED THE NUMBER OF PEOPLE IN MY VICINITY (SO THE # OF PEOPLE IN THE ROOM I WAS IN, OR THE # OF PEOPLE IN ANY ENCLOSED SPACE). THIS WAS HARD TO GATHER, HENCE ME RESORTING TO REPRESENTING THE DATA THROUGH USING BANDING. NOTE: SOME DATA IS MISSING DUE TO MY HAPHAZARD DATA GATHERING!

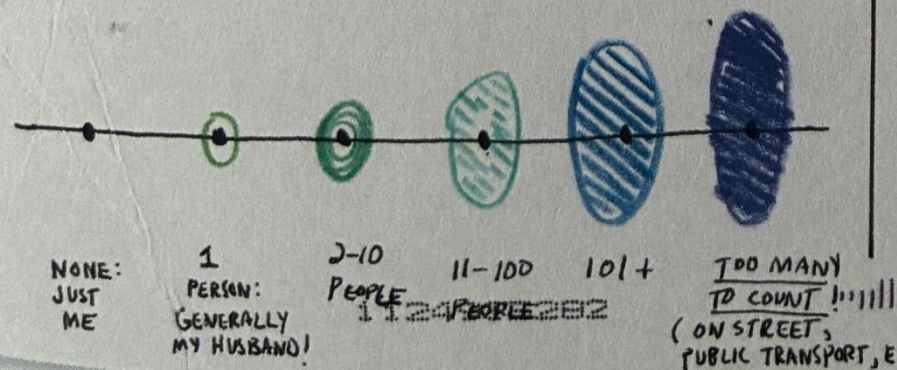
HOW TO READ IT:

(NOT MUCH SLEEP THIS WEEK)

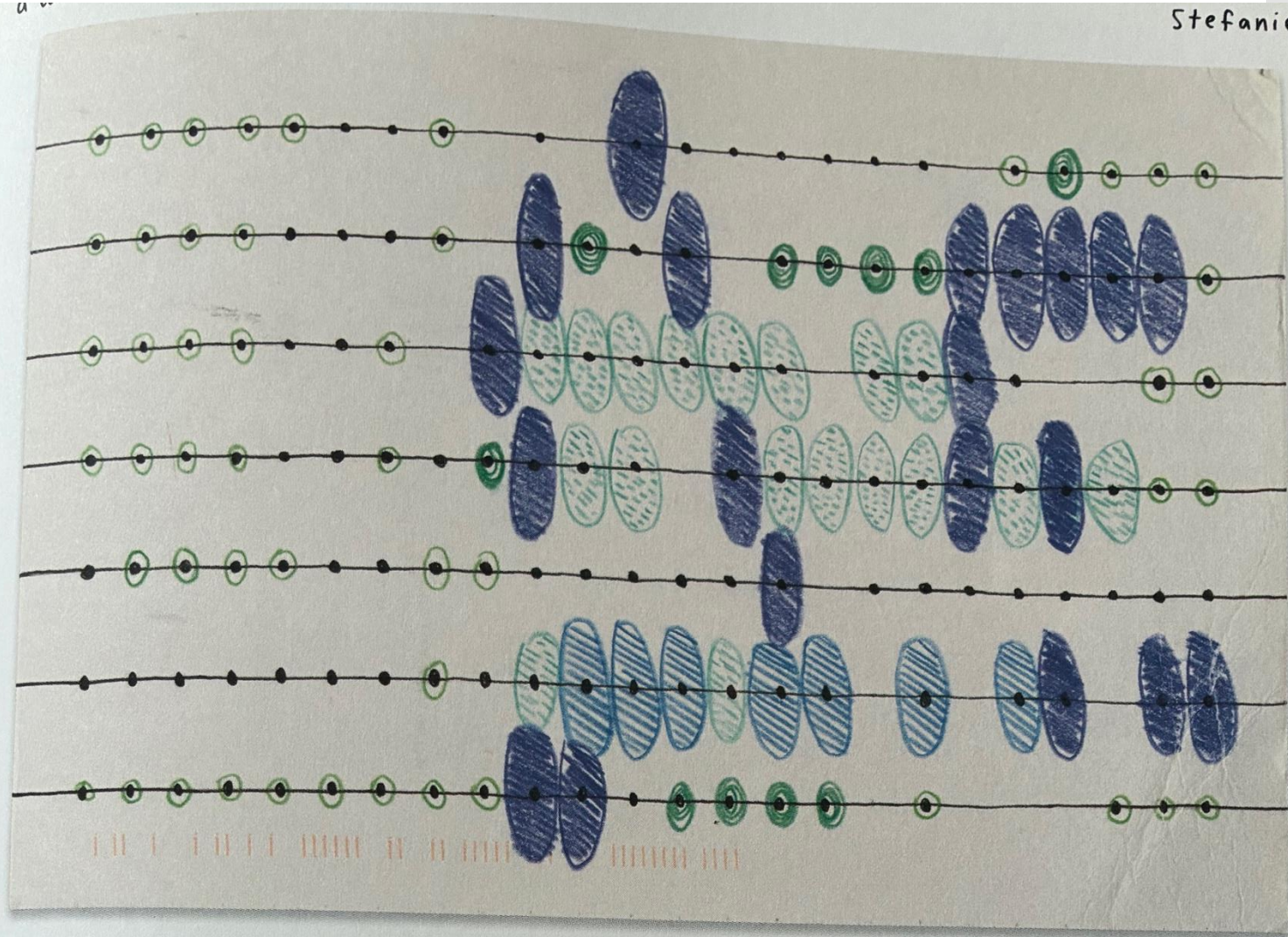


BLACK DOTS EACH REPRESENT AN HOUR WHERE THERE IS DATA.

NUMBER OF PEOPLE IN MY SPACE/VICINITY:



Stefanie



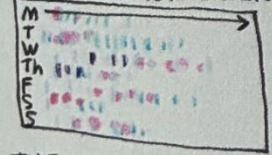
DEAR DATA - WEEK 24

A WEEK OF DOORS/SPACES

ABOUT THE DATA: I GATHERED DATA ON ALL OF THE SPACES I PASSED THROUGH IN THE WEEK, BOTH INTERNAL + EXTERNAL. A SPACE IS DEFINED BY WHETHER I HAD TO PASS THROUGH A DOOR OR NOT.

HOW TO READ IT:

IN CHRONOLOGICAL ORDER



EACH SPACE IS REPRESENTED BY A:

OR A (CROSS-HATCHED LINES CARRY NO IMPORTANCE, MAINLY ONLY FOR DIFFERENTIATION BETWEEN SIMILAR COLOURS) THIS

THE TYPES OF SPACES I PASSED THROUGH INCLUDE: WAS A LAST MINUTE SPONTANEOUS DECISION ON THE 1ST CARD TO MAKE THE COLOURS CLEARER... SOOO ANNOYING I HAVE TO REDRAW

IN THE HOME/

PERSONAL SPACE:

+++++ BEDROOM
SPARE BEDROOM (1 OF 2)
GROUND FLOOR (OPEN PLAN, INCLS. KITCHEN + LOUNGE)
TOILET
BATHROOM

STUDIO:

+++++ STUDIO BLDG.
STUDIO SPACE

OUTSIDE:

IN THE STREET

DOCTOR:

DOCTOR'S OFFICE
WAITING ROOM

UNIVERSITIES:

UNIV. BLDG.
CLASSROOM
TEAROOM

CULTURAL CENTRE:

MAIN BLDG.
GALLERY

TRANSPORT:

TUBE/TRAIN
TUBE/TRAIN STN
CAB

SHOP

BRYXTON MKT. BLDG.
PUB/RESTAURANT/CAFE/BAR

FRIEND'S HOUSE:

MAIN HOUSE
BASEMENT
KITCHEN
LOUNGE

GYM:

MAIN BLDG.
CHANGING ROOM
EXERCISE ROOM

* CHANGING ROOMS + TOILETS ARE FOUND ACROSS DIFFERENT SPACES

FROM:

S. POSAVEC

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To:

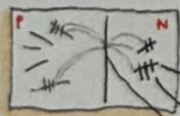
LEFT-HAND SIDE
SMUDGES

Stefanie

DEAR DATA

WEEK 29: MY BOYFRIEND

HOW TO READ IT?



P = positive ones
N = negative ones

This week I noticed and recorded everything that my boyfriend did that caused any emotional reaction on me.

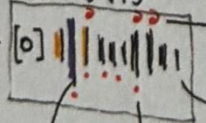
Each group represents a type of action
Each single bar is the actual action, in chronological order

CONNECTIONS = things that can likely happen in a regular week = always

[.] SYMBOL = TYPOLOGY

- P • helped me physically
- helped me in general
- * took special care of me
- ♥ showed me love
- reassured me
- △ put my needs before his ones
- △ let me do weird stuff
- * figured out stuff I couldn't
- showed me how wise and BETTER he is!

ELEMENTS



red dot above = if present it means I said THANK you 😊
|| = "intensity of the action"

COLOR = context

- work related
- Dear data rel.
- personal stuff
- life/practical stuff
- opinions
- objects/physical stuff
- the 2 of us

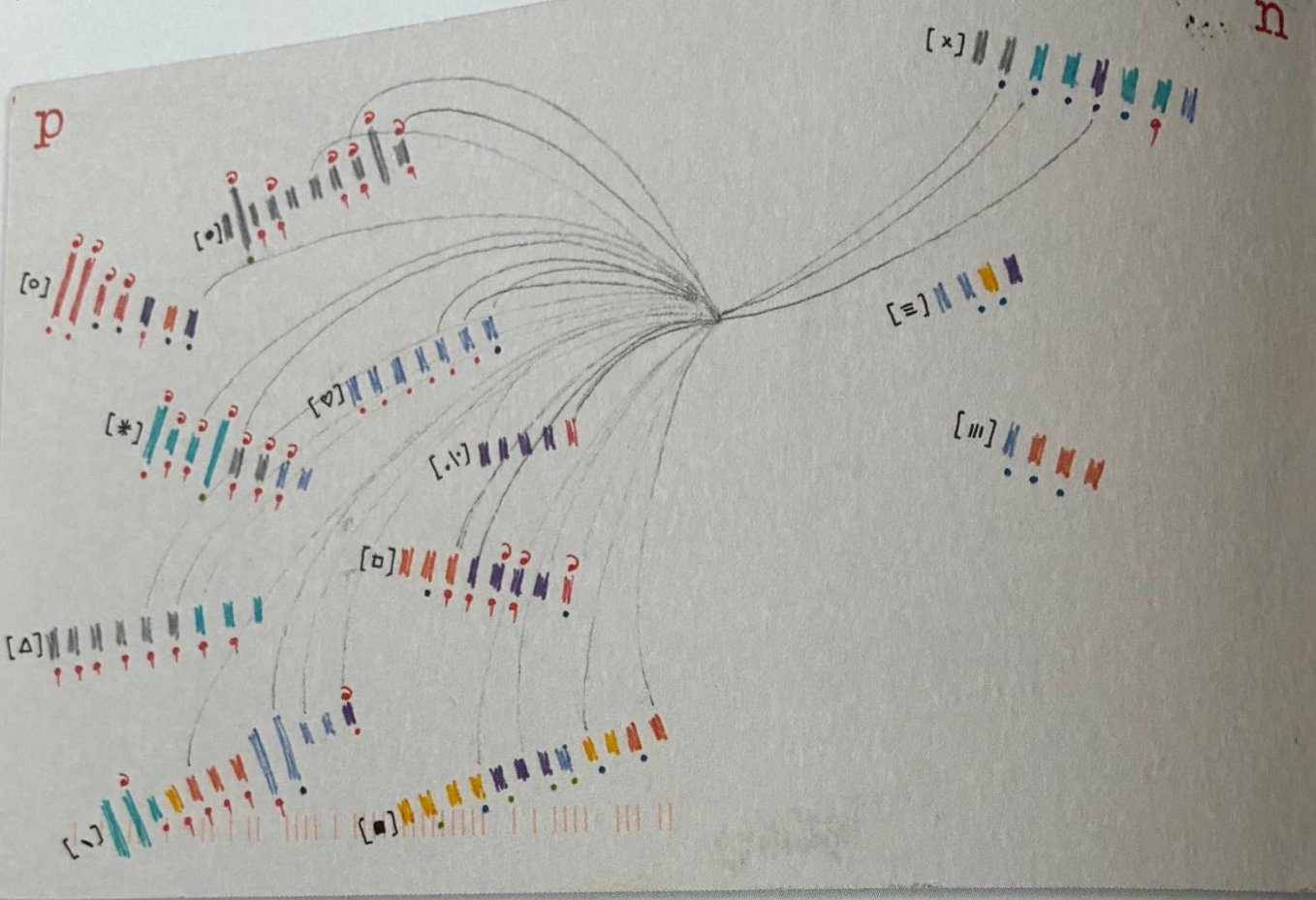
|| = regular
|| = very special thing

DOT BELOW = my reaction!

- surprised
- pleased
- felt better!
- annoyed
- imitated

void = nothing particular
wouldn't have noticed that for this Dear DATA week

GEORGIA



Georgia found this an incredibly partner trigger

Your Turn

Find a small group to work with.

1. Pick a topic of interest and find data related to it.
2. Select two specific variables.
3. Sketch two different visual encodings of the variables. One should be simple, and one should be more creative/complex.